

Existence and Uniqueness of Correct Steady States in Discrete-Time Closed Loops Under Adversarial Attacks

Executive Summary

A discrete-time “steady state” is, most fundamentally, a fixed point of a time-invariant closed-loop map T : an x^* satisfying $x^* = T(x^*)$. Under adversarial attacks, the closed loop typically becomes time-varying (a family $\tilde{T}_k$), and a single fixed point may not exist unless the attack converges to a constant (or is itself constant). In that more general case, the mathematically correct “steady-state object” is usually a *robust positively invariant* (RPI) set or an *ultimate bound*, rather than a point equilibrium.

Existence of equilibria in attacked discrete-time systems can be guaranteed under several alternative assumption sets. The main clean routes are:

- (i) Brouwer existence when T is continuous and maps a compact convex invariant set into itself;
- (ii) Schauder existence in infinite-dimensional/compact-operator settings;
- (iii) Kakutani existence for discontinuous/variable-structure dynamics modeled as a set-valued map (relevant when signum or switching is present).

Uniqueness of the equilibrium (and global convergence to it) is most directly obtained via Banach’s contraction principle, i.e., a global Lipschitz constant $L < 1$ of the closed-loop map; this also yields explicit robustness bounds and “thresholds” of attack magnitude beyond which invariance or contraction fails.

Regarding paper [1]: it designs a tracking controller/neural-network-like structure for a specified class of discrete-time nonlinear systems that are affine in the input and whose internal dynamics $f(\cdot)$ and input coupling $g(\cdot)$ are assumed known. Its derived control law depends explicitly on $g(x(k))^{-1}$ (or a special case when $g(x(k)) = 0$), and its tracking/error dynamics use a signum nonlinearity. Therefore, it does *not* claim unconditional “return to equilibrium regardless of how the system is modified.” Its claims are conditional on the model class and assumptions under which the control equation is derived.

Finally, because you asked to align with paper [3] as a close analog but [3] is not explicitly provided here, this report treats [3] as unspecified and analyzes several plausible options (continuous vs. discrete time; signum-based variable structure vs. smooth gradient descent). Where helpful, it also points to a highly plausible candidate analog in Pavlo Tymoshchuk’s closely related continuous-time formulation (same control-affine structure and signum-based error dynamics).

1 Problem Setup and Attacked Closed-Loop Difference Equations

This section delivers: nominal closed-loop equation(s) and attacked variants for attacks on state dynamics, control channel, reference, and measurements.

1.1 Nominal Plant, Sensing, and Controller

Let $x_k \in \mathbb{R}^n$ be the state, $u_k \in \mathbb{R}^m$ the control, $y_k \in \mathbb{R}^p$ the measured output, and $r_k \in \mathbb{R}^q$ the reference. A general discrete-time plant and sensor are

$$x_{k+1} = F_k(x_k, u_k), \quad y_k = H_k(x_k). \quad (1)$$

A (possibly dynamic) controller is

$$z_{k+1} = \phi_k(z_k, y_k, r_k), \quad u_k = \psi_k(z_k, y_k, r_k), \quad (2)$$

so the closed-loop state is $\xi_k := (x_k, z_k)$ and its evolution is a difference equation $\xi_{k+1} = T_k(\xi_k)$.

When time-invariant and the reference is constant ($r_k \equiv r$), the closed loop is autonomous:

$$\xi_{k+1} = T(\xi_k). \quad (3)$$

1.2 Model Aligned with Paper [1] and a Plausible Paper [3] Analog

Paper [1] uses an Euler-discretized control-affine nonlinear system:

$$x(k+1) = x(k) + \tau(f(x(k)) + g(x(k))u(k)). \quad (4)$$

A closely related continuous-time analog (a strong candidate for “paper [3]” if [3] is a predecessor/analog) uses

$$\dot{x}(t) = f(x(t)) + g(x(t))u(t), \quad (5)$$

together with explicit assumptions such as Lipschitz continuity on a set and a controllability/nondegeneracy condition $g(x) \neq 0$ (and a special-case treatment when $g(x) = 0$).

To keep the rest of this report discrete-time (as requested), we define the nominal closed-loop map for constant reference r as

$$x_{k+1} = T(x_k) := F(x_k, \kappa(H(x_k), r)), \quad (6)$$

or for augmented state $\xi = (x, z)$,

$$\xi_{k+1} = T(\xi_k). \quad (7)$$

1.3 Attacked Variants

Cyber-physical security literature commonly models attacks as exogenous “unknown inputs” corrupting actuation and sensing channels. Define the attack signals:

- $a_k^x \in \mathbb{R}^n$: state-dynamics (process) injection,
- $a_k^u \in \mathbb{R}^m$: actuator/control-channel injection,
- $a_k^y \in \mathbb{R}^p$: sensor/measurement falsification,
- $a_k^r \in \mathbb{R}^q$: reference corruption,
- a_k^θ : structured parametric/model perturbation.

Then the attacked system variants are:

State/process attack.

$$x_{k+1} = F(x_k, u_k) + a_k^x. \quad (8)$$

Actuator attack.

$$\tilde{u}_k = u_k + a_k^u, \quad x_{k+1} = F(x_k, \tilde{u}_k). \quad (9)$$

Sensor attack.

$$\tilde{y}_k = H(x_k) + a_k^y, \quad u_k = \kappa(\tilde{y}_k, r_k) \text{ (or using a controller state } z_k). \quad (10)$$

Reference attack.

$$\tilde{r}_k = r_k + a_k^r, \quad u_k = \kappa(y_k, \tilde{r}_k). \quad (11)$$

Structured parametric (model) attack.

$$x_{k+1} = F(x_k, u_k; \theta + a_k^\theta). \quad (12)$$

These can be summarized as a time-varying attacked closed-loop map

$$x_{k+1} = \tilde{T}_k(x_k), \quad (13)$$

or $\xi_{k+1} = \tilde{T}_k(\xi_k)$, where $\tilde{T}_k$ depends on which signals the adversary can modify.

2 Admissible Attack Classes and Assumption Sets

2.1 Admissible Attack Classes

If the paper does not explicitly constrain attacks, then the correct interpretation is: *no constraint*. Under no constraint, existence/uniqueness of a “correct” steady state is generally not provable because the map $\tilde{T}_k$ can be arbitrarily altered by an adversary at each time step. This “limitations without assumptions” perspective is explicit in system-theoretic security work that characterizes fundamental monitoring/detection limitations under an omniscient adversary model.

To enable rigorous equilibrium analysis, introduce one (or more) admissible classes:

2.2 Assumption Sets for Existence and Uniqueness

Because the system/controller class is not fully specified, the best paper-ready approach is to declare alternative assumption sets and clearly state what each yields.

Assumption set A: continuous self-map on a compact convex invariant set (Brouwer existence). There exists a nonempty compact convex set $K \subset \mathbb{R}^n$ such that:

- (a) $T : K \rightarrow K$ is continuous, and
- (b) $T(K) \subseteq K$.

Assumption set B: contraction of the closed-loop map (Banach existence + uniqueness). There exists a norm $\|\cdot\|$ and constant $L \in [0, 1)$ such that

$$\|T(x) - T(y)\| \leq L \|x - y\| \quad \forall x, y. \quad (14)$$

Then a unique fixed point exists and all trajectories converge to it.

Attack class	Formal model	Key equilibrium implication	Primary sources
No constraint (omniscient)	a_k arbitrary, possibly state-dependent and unbounded	No general existence/uniqueness guarantees	Pasqualetti–Dörfler–Bullo
Bounded additive	$\ a_k^x\ \leq \bar{a}_x$, $\ a_k^u\ \leq \bar{a}_u$, etc.	Often implies boundedness/ultimate bounds; fixed point only if attack converges to constant	ISS (Sontag)
Persistent constant (bias)	$a_k \equiv a$ constant	Produces a shifted equilibrium solving $x = T(x) + d$	Bias/stealthy injection literature
Sparse/occasional	$a_k \neq 0$ only on a sparse time set; or only s sensors corrupted	If nominal map is contracting, trajectories recover after attacks stop	Fawzi–Tabuada–Diggavi
Structured parametric	$\theta_k \in \Theta$ (bounded set)	Existence may persist; uniqueness can fail	Teixeira–Sandberg–Johansson
Worst-case under stealth constraints	Attacker chooses a_k subject to undetectability	Even bounded attacks can be strategically destabilizing	Stealth attack literature

Table 1: Admissible attack classes and their equilibrium implications.

Assumption set C: monotonicity/strong monotonicity of the equilibrium operator (uniqueness without global contraction). Define the equilibrium operator $A(x) := x - T(x)$. If A is strongly monotone (or strictly monotone in scalar cases), then $A(x) = d$ has at most one solution, yielding uniqueness of the fixed point under constant bias. This connects to monotone operator theory (Minty, Rockafellar).

Assumption set D: discontinuities modeled as set-valued maps (Kakutani existence). If T contains discontinuities (e.g., signum/switching as in paper [1]), replace it with a set-valued correspondence $\mathcal{T}$ (e.g., using $\text{sgn}(0) \in [-1, 1]$ componentwise). Under standard convexity and upper-hemicontinuity/closed-graph conditions, Kakutani yields existence of $x^* \in \mathcal{T}(x^*)$.

Assumption set E: Lyapunov/LaSalle invariance for discrete-time (convergence to invariant sets under disturbances). If attacks are bounded but time-varying, equilibrium may not exist; instead show convergence to an invariant set using discrete-time Lyapunov and LaSalle-type arguments (Mei–Bullo).

3 Steady-State Definitions and Fixed-Point Equations Under Attacks

3.1 What “Correct Steady State” Can Mean

One should pick (and state) one of the following, because different notions lead to different theorems:

Definition 3.1 (Reference-consistent equilibrium (stringent)). *For constant r , the “correct equilibrium” is $x^* = r$ (or error $e^* = 0$), and one requires that x^* remains a fixed point even under attack.*

Definition 3.2 (Robust equilibrium (practical)). *The “correct equilibrium” is the unique equilibrium of the attacked closed loop (which may shift with bias) and is close to the nominal one when the attack is small.*

Remark 3.3. *For time-varying attacks, it is often more correct to define the “steady state” as a robust invariant set or an ultimate bound rather than a point.*

3.2 Fixed-Point Equations for Constant Attacks

Assume time-invariance and constant reference r , and consider constant (bias-type) attacks so the attacked map is time-invariant.

Nominal equilibrium.

$$x^* = T(x^*) = F(x^*, \kappa(H(x^*), r)). \quad (15)$$

State/process bias ($a_k^x \equiv d_x$).

$$x^* = T(x^*) + d_x. \quad (16)$$

Actuator bias ($a_k^u \equiv d_u$).

$$x^* = F(x^*, \kappa(H(x^*), r) + d_u). \quad (17)$$

Sensor bias ($a_k^y \equiv d_y$).

$$x^* = F(x^*, \kappa(H(x^*) + d_y, r)). \quad (18)$$

Reference bias ($a_k^r \equiv d_r$).

$$x^* = F(x^*, \kappa(H(x^*), r + d_r)). \quad (19)$$

Structured parametric bias ($\theta \mapsto \theta + d_\theta$).

$$x^* = T_{\theta+d_\theta}(x^*). \quad (20)$$

3.3 What Changes for Time-Varying Attacks

If any attack is time-varying with no limit (e.g., a_k oscillatory), then the equilibrium equation is not well-defined because the map is $\tilde{T}_k$, not a single T .

In that case, the right “steady-state” concepts are:

- (a) a bounded invariant set X such that $\tilde{T}_k(X) \subseteq X$ for all k , or
- (b) an ultimate bound $\limsup_{k \rightarrow \infty} \text{dist}(x_k, X) \leq \epsilon$, or
- (c) an invariant trajectory $x_{k+1}^* = \tilde{T}_k(x_k^*)$ (rarely closed-form).

RPI sets for disturbed discrete-time linear systems are treated in the disturbance-invariant set literature.

4 Existence Results with Paper-Ready Theorem Statements

4.1 Existence via Brouwer for Continuous Maps

Theorem 4.1 (Existence of an equilibrium on a compact convex invariant set). *Let $K \subset \mathbb{R}^n$ be nonempty, compact, and convex. Let $T : K \rightarrow K$ be continuous. Then there exists $x^* \in K$ such that $x^* = T(x^*)$.*

Corollary 4.2 (Existence under constant additive attack). *If $T_d(x) := T(x) + d$ maps K into itself (i.e., $T_d(K) \subseteq K$), then there exists $x_d^* \in K$ such that $x_d^* = T(x_d^*) + d$.*

Paper-inclusion proof sketch (compactness/invariance route). Choose a compact convex K that is forward invariant; apply Brouwer to T (or T_d) restricted to K .

How to make this usable under attacks. The technical work becomes: construct K such that, for every admissible constant attack in the class, the attacked map remains a self-map on K . This directly produces a quantitative attack-magnitude threshold.

4.2 Existence via Schauder for Compact Operators

If the closed-loop operator is compact (maps bounded sets to relatively compact sets) on a closed convex set in a Banach space, Schauder yields existence:

Theorem 4.3 (Schauder fixed point; compact-image version). *Let C be a nonempty, closed, convex subset of a Banach space. If $T : C \rightarrow C$ is continuous and $T(C)$ is compact, then T has a fixed point.*

This is most relevant if the “state-variable trajectory” is treated as an element of a function space (e.g., proving existence of a periodic orbit of a forced system via a Poincaré map, or analyzing an operator on bounded sequences).

4.3 Existence for Signum/Switching via Kakutani

Paper [1] explicitly uses signum in the tracking error dynamics and the deployed tracking equation. A signum term is discontinuous, so Brouwer/Schauder continuity hypotheses can fail. A standard remedy is to interpret the dynamics as a set-valued map at the discontinuity.

Construction. Replace $\text{sgn}(0)$ with the convex set $[-1, 1]$ (componentwise), forming a correspondence $\mathcal{T}$.

Theorem 4.4 (Kakutani fixed point; set-valued existence). *Let $K \subset \mathbb{R}^n$ be nonempty, compact, convex. Let $\mathcal{T} : K \rightrightarrows K$ have nonempty, convex, closed values and satisfy the closed-graph (upper-hemicontinuity) property. Then there exists $x^* \in K$ such that $x^* \in \mathcal{T}(x^*)$.*

This theorem is precisely what allows one to state existence of an equilibrium even when the closed-loop “map” includes switching or signum.

4.4 Existence via Invariance Under Bounded Disturbances (Set-Based Steady State)

For bounded time-varying attacks, a point equilibrium may not exist, but an invariant set does. For discrete-time linear systems $x_{k+1} = Ax_k + w_k$ with $w_k \in W$ (compact), the disturbance-invariant set framework provides existence (and computation/characterization) of invariant sets inside a constraint set Γ . This is treated in detail by Kolmanovsky and Gilbert.

5 Uniqueness and Robustness Thresholds

5.1 Uniqueness via Banach Contraction

Theorem 5.1 (Banach contraction principle). *If T is a contraction on a complete metric space (X, d) , then T has a unique fixed point and iterates converge to it from any initial condition.*

Attack-robust corollaries (paper-ready bounds).

Corollary 5.2 (Constant additive attacks preserve uniqueness). *If T is a contraction with constant $L < 1$, then $T_d(x) = T(x) + d$ is also a contraction with the same L ; hence there is a unique attacked equilibrium x_d^* .*

Corollary 5.3 (Equilibrium shift bound / bias-to-state gain). *Let x^* be the nominal fixed point and x_d^* the fixed point under constant additive disturbance d . Then*

$$\|x_d^* - x^*\| \leq \frac{\|d\|}{1-L}. \quad (21)$$

Corollary 5.4 (Bounded time-varying disturbance gives an ultimate bound). *For $x_{k+1} = T(x_k) + w_k$ with $\|w_k\| \leq \bar{w}$ and T contractive with constant L , one obtains*

$$\limsup_{k \rightarrow \infty} \|x_k - x^*\| \leq \frac{\bar{w}}{1-L}. \quad (22)$$

This is a discrete-time ISS-type bound, aligning with the ISS philosophy formalized by Sontag.

Threshold interpretation. The critical “existence/uniqueness threshold” under structured parametric attacks is typically the boundary where the contraction property is lost (effective $L \geq 1$), or—in linear systems—the boundary where the spectral radius crosses 1.

5.2 Uniqueness via Strict/Strong Monotonicity

Consider the equilibrium equation under constant additive attack:

$$x = T(x) + d \iff A(x) := x - T(x) = d. \quad (23)$$

If A is strongly monotone, then $A(x) = d$ has at most one solution (hence the attacked equilibrium is unique). Strong monotonicity and maximal monotonicity are central in monotone operator theory, with foundational results by Minty and Rockafellar.

This route is especially natural if the controller can be written as a proximal/gradient-type recursion, where A becomes a gradient of a strongly convex function (then uniqueness is immediate).

5.3 Uniqueness/Convergence via Lyapunov–LaSalle (Discrete Time)

If one can produce a Lyapunov function V such that the nominal closed loop satisfies

$$V(x_{k+1}) - V(x_k) \leq -W(x_k) \quad (24)$$

for some positive definite W , one can prove convergence to the largest invariant set where $W = 0$. For discrete-time systems, a careful, self-contained LaSalle development is given by Mei and Bullo.

Under bounded disturbances (attacks), Lyapunov methods typically yield convergence to a neighborhood/invariant set whose size scales with the disturbance bound—again matching the ISS perspective.

5.4 When Guarantees Fail

A compact way to state failure modes:

- Without an invariant set K , Brouwer/Schauder routes do not apply.
- Without contraction (or strong monotonicity), uniqueness need not hold (multiple equilibria or limit cycles may appear).

- Under time-varying attacks with no convergence to a constant, point equilibria are generally not the right object; expect invariant sets or oscillations instead.
- Under unconstrained omniscient adversaries, stability and even detectability can fail; explicit attack constraints are essential for meaningful guarantees.

6 Comparison to Paper [1] and Alignment Choices for Paper [3]

6.1 What Paper [1] Actually Assumes and Claims

Paper [1] (as available in the cited proceedings PDF) frames tracking as: design a system whose output approximately repeats a predefined time-varying reference trajectory.

It adopts a time-invariant, control-affine discrete-time nonlinear system (via Euler approximation):

$$x(k+1) = x(k) + \tau(f(x(k)) + g(x(k))u(k)), \quad (25)$$

and explicitly treats $f(\cdot)$ and $g(\cdot)$ as known internal dynamics/coupling maps.

A key structural dependence is that the control equation is written using $g(x(k))^{-1}$ and a special fallback when $g(x(k)) = 0$. This alone prevents any interpretation like “no matter how the system is modified”: if g changes (or becomes singular/degenerate), the controller derived for the original g is no longer the controller for the modified system.

Paper [1] also introduces closed-loop tracking-error dynamics with a signum nonlinearity:

$$e(k+1) = e(k) - \beta \operatorname{sgn}(e(k)), \quad (26)$$

and gives an equivalent deployed tracking equation for $x(k)$, emphasizing that the tracking equation describes the discrete-time state trajectory of the designed system.

Finally, paper [1] explicitly positions itself as an extension of a prior continuous-time method and attributes to that earlier method that the control “provides finite convergence time of the system state trajectories to steady states” and allows changing convergence time via a learning-rate parameter. Those statements are *not* universal robustness claims; they are method statements under the paper’s modeling and structural assumptions.

Therefore: paper [1] should be characterized as a tracking-control design for a specified model class (control-affine, known dynamics) with a particular variable-structure (signum-based) tracking-error recursion. It does *not* claim unconditional return-to-equilibrium under arbitrary system modification.

6.2 How the Adversarial Formulation Tightens/Extends Paper [1]

Paper [1] does not model an explicit adversary in its difference equations. The adversarial extension forces two clarifications:

1. **What attacks do to “correct equilibrium.”** A constant bias attack generally shifts equilibria unless the controller includes disturbance rejection tailored to that class (e.g., integral action or an estimator). In general, “correct = $(x = r)$ ” is not preserved under bias.
2. **What steady-state object exists under time-varying attacks.** Time-varying attacks typically require invariant-set/ISS analysis rather than fixed-point analysis.

6.3 Paper [3] Is Unspecified: Plausible Variants and Their Impact

Because paper [3] is not explicitly provided here, the robust way to proceed is to state “alignment options.”

Option 1: [3] is a continuous-time analog with signum-based variable structure (likely). A closely related continuous-time control-affine system is presented in a companion-like text with assumptions including Lipschitz continuity on a set, stabilizability, known (f, g) , and $g(x) \neq 0$ (or a special case when $g(x) = 0$). *Impact:* to move to discrete time, existence/uniqueness proofs become fixed-point/invariance proofs for the discretized map. Discontinuities from signum motivate Kakutani in discrete-time, while continuous-time would require Filippov/differential inclusion machinery.

Option 2: [3] replaces signum with smooth gradient descent. If [3] uses a smooth gradient step (e.g., $e_{k+1} = e_k - \alpha \nabla C(e_k)$), then T is typically continuous (often Lipschitz). *Impact:* Brouwer/Banach become directly applicable; set-valued analysis may not be needed.

Option 3: [3] uses unknown internal dynamics and adaptive/approximate elements. Some of Tymoshchuk’s later work streams discuss unknown dynamics and emphasize existence/uniqueness of steady states in that context. *Impact:* one must separate (i) nominal/true plant map from (ii) implemented controller map (with estimation/approximation error). Existence/uniqueness then becomes a robustness problem: does approximation error preserve contraction/invariance?

If paper [3] is provided later: the sections that would need revision are the “alignment option” assumptions (continuity vs. set-valued; known vs. unknown (f, g) ; continuous vs. discrete time). The general attack modeling, fixed-point equations, and the example templates remain structurally the same.

7 Worked Examples with Analytic Derivations

7.1 Linear Affine-in-Input Example

Nominal plant and controller. Let

$$x_{k+1} = a x_k + b u_k, \quad r_k \equiv r. \quad (27)$$

Choose a tracking controller

$$u_k = -K(x_k - r) + u_{\text{eq}}, \quad u_{\text{eq}} = \frac{(1-a)}{b} r, \quad (28)$$

so the nominal closed-loop is

$$x_{k+1} = \alpha x_k + (1-\alpha) r, \quad \alpha = a - bK. \quad (29)$$

If $|\alpha| < 1$, the map is a contraction in $|\cdot|$ and has the unique fixed point $x^* = r$ (Banach route).

Constant actuator bias attack. Let $\tilde{u}_k = u_k + a^u$ with constant a^u . Then

$$x_{k+1} = \alpha x_k + (1-\alpha) r + b a^u, \quad (30)$$

and the unique equilibrium is shifted to

$$x_{a^u}^* = r + \frac{b a^u}{1-\alpha}. \quad (31)$$

This is exactly the fixed-point equation $x = T(x) + d$ with a contraction T , hence unique.

Time-varying bounded actuator attacks. If a_k^u is bounded but time-varying, there may be no equilibrium; instead, the state remains bounded and approaches an ultimate bound of size proportional to $\sup_k |a_k^u|$ when $|\alpha| < 1$, consistent with ISS intuition.

Threshold in this example. Uniqueness and global convergence hold when $|\alpha| < 1$. A structured parametric attack that changes a or K can force $|\alpha| \geq 1$, at which point contraction and convergence can fail.

7.2 Nonlinear Example with Uniqueness–Multistability Threshold

This example shows that uniqueness can fail (multiple equilibria) even when equilibria exist, and that a parameter threshold (which an attacker could perturb) determines whether uniqueness holds.

Plant and feedback. Let

$$x_{k+1} = x_k + \tau(f(x_k) + u_k) + w_k, \quad f(x) = x - x^3, \quad (32)$$

and let $u_k = -k(x_k - r)$ with constant r . Then

$$x_{k+1} = x_k + \tau(kr + (1 - k)x_k - x_k^3) + w_k. \quad (33)$$

Constant disturbance (bias) equilibrium equation. If $w_k \equiv w$, equilibria satisfy

$$x^* = x^* + \tau(kr + (1 - k)x^* - (x^*)^3) + w \iff (x^*)^3 + (k - 1)x^* - kr = \frac{w}{\tau}. \quad (34)$$

A real solution always exists because the left side is continuous and unbounded as $|x^*| \rightarrow \infty$.

Uniqueness condition via strict monotonicity. Define $\varphi(x) = x^3 + (k - 1)x - kr - \frac{w}{\tau}$. Then $\varphi'(x) = 3x^2 + (k - 1)$.

- If $k > 1$, then $\varphi'(x) > 0$ for all x , so φ is strictly increasing and the equilibrium is unique for every (r, w) .
- If $k < 1$, φ' can be negative near 0, and the cubic can have three real roots (multiple equilibria), so uniqueness fails for some (r, w) .

A sharp multistability condition can be written using the depressed cubic discriminant. For $x^3 + px + q = 0$ with $p = k - 1 < 0$, $q = -kr - \frac{w}{\tau}$, there are three distinct real roots if and only if $27q^2 < -4p^3$, i.e.,

$$\left| kr + \frac{w}{\tau} \right| < \frac{2}{3\sqrt{3}} (1 - k)^{3/2}. \quad (35)$$

This gives an explicit “attack/parameter threshold”: if an attacker can change k downward (structured parametric attack), it can create multiple equilibria.

8 Summary Table

9 Recommended Citations and Paper-Ready Blocks

9.1 Primary Sources

1. Fixed-point existence (finite-dimensional): L. E. J. Brouwer (1912).
2. Contraction/uniqueness: Stefan Banach (1922).

Attack class	Time var.	Best “steady-state” claim	Existence (sufficient)	Uniqueness (sufficient)	Failure threshold
Unconstrained omniscient	arbitrary	No general guarantee	Not provable w/o extra structure	Not provable	Adversary breaks invariance
Constant bounded bias	constant	Point equilibrium (possibly shifted)	Brouwer: cont. self-map on compact convex K	Banach: $L < 1$	Invariance margin violated; $L \uparrow 1$
Bounded time-varying	bounded	Invariant set / ultimate bound (ISS/RPI)	RPI set existence	Minimal RPI (problem-specific)	Bound grows $\propto 1/(1 - L)$
Sparse / occasional	intermittent	Recovery if nominal stable + attacks stop	Contraction or Lyapunov + finite attack horizon	Same as nominal	Persistence constraints violated
Structured parametric	bounded set	Equilibrium may bifurcate	Continuity + invariance over parameter set	Contraction/monotonicity over set	Parameter crosses stability boundary

Table 2: Summary of existence and uniqueness guarantees by attack class.

3. Infinite-dimensional fixed points: Juliusz Schauder (1930).
4. Set-valued fixed points for discontinuities: Shizuo Kakutani (1941).
5. Security/attack modeling as unknown inputs; limitations: Pasqualetti–Dörfler–Bullo.
6. Resource-limited adversary attack spaces and safe sets: Teixeira–Shames–Sandberg–Johansson.
7. False-data injection in LQG/Kalman settings: Yilin Mo and Bruno Sinopoli.
8. Sparse sensor/actuator corruption: Fawzi–Tabuada–Diggavi.
9. Disturbance invariant sets (discrete time): Kolmanovsky–Gilbert.
10. Discrete-time LaSalle invariance principle toolkit: Mei–Bullo.
11. Monotone operator uniqueness route: Minty and Rockafellar.

9.2 Paper-Ready Theorem/Lemma Blocks

Lemma 9.1 (Attacked fixed-point equation). *If $x_{k+1} = T(x_k) + d$ with constant d , then equilibria satisfy $x^* = T(x^*) + d$.*

Theorem 9.2 (Brouwer existence under invariant compact convex set). *If $T : K \rightarrow K$ is continuous and K is nonempty compact convex, then T has a fixed point.*

Theorem 9.3 (Banach uniqueness and convergence). *If T is a contraction with constant $L < 1$, then it has a unique fixed point and $x_k \rightarrow x^*$ for all initial conditions.*

Corollary 9.4 (Bias-to-equilibrium bound). *If T is contractive with constant $L < 1$, then $\|x_d^* - x^*\| \leq \|d\|/(1 - L)$.*

Theorem 9.5 (Kakutani existence for signum/switching). *If $\mathcal{T}$ is a Kakutani correspondence on compact convex K , then $\exists x^* \in K$ with $x^* \in \mathcal{T}(x^*)$.*

Theorem 9.6 (Discrete-time LaSalle invariance principle). *Use a discrete-time LaSalle statement suitable for the $V(x_{k+1}) - V(x_k) \leq 0$ structure and characterize the limit set as the largest invariant set inside $\{x : \Delta V(x) = 0\}$.*